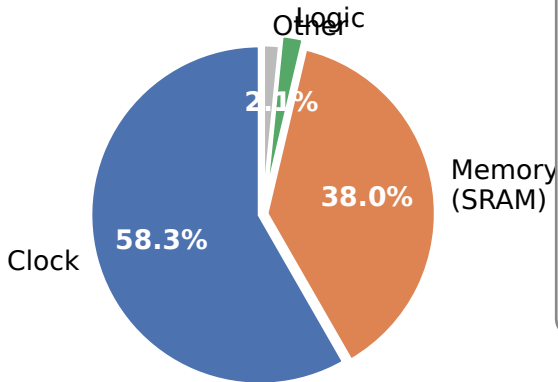


(a) Baseline dynamic power
(post-synthesis, TSMC 28nm)



(b) Implication

Key observation

- Data movement + clock account for 96.3% of power
- Compute logic is only 2.1%

Horowitz, ISSCC'14 [6]:

$$E_{move} / E_{compute} \approx 10-100\times$$

→ Optimizing memory access yields the largest energy win

Fig. 1. Baseline power breakdown motivates memory-centric optimization.